MECH 467 Project III: with Pole Placement Controller

MARIANNE KOBIERSKI
STUDENT ID: 27475110
KOBIERSKI.M@ALUMNI.UBC.CA
LAB PREFORMED ON 28 NOV 2014

1 ABSTRACT

The objective of this report is to discuss the design and behaviour of a pole placement controller for use in the same situation described in Lab 3.

2 INTRODUCTION

This report expands on Lab 3. A pole placement controller was designed and tested on Virtual CNC. The simulated behaviour of the pole placement controller is presented here, with observations contrasting the behaviour of a low-pass filter in comparison, in terms of tracking and contouring error.

3 Pole Placement Controller Toolpath and Dynamics Profiles

The pole placement controller (PPC) was simulated using the Virtual CNC software, with the same desired trace as was used for the three controllers in Lab 3. The desired and simulated toolpaths are shown in Figure 1.

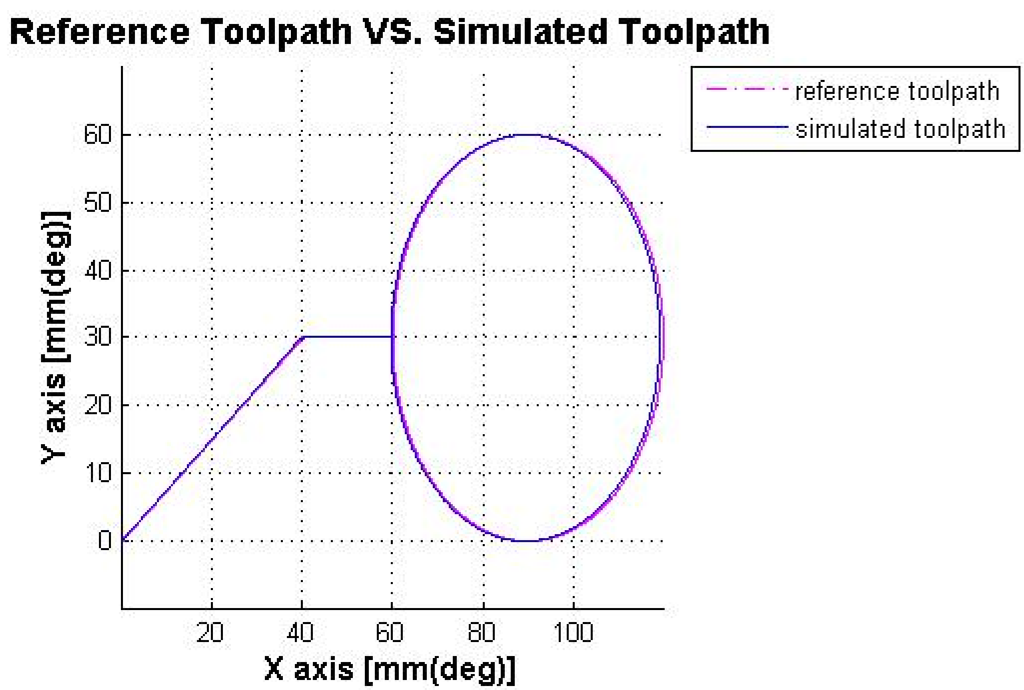

Figure 1: Reference and Simulated Toolpaths for PPC

There appears to be good agreement along the Y axis, but some discrepancies in positioning along the X axis.

To better view these discrepancies, the X and Y positions, velocities, and accelerations were plotted for the ideal and simulated situations (Figures 2 – 4).

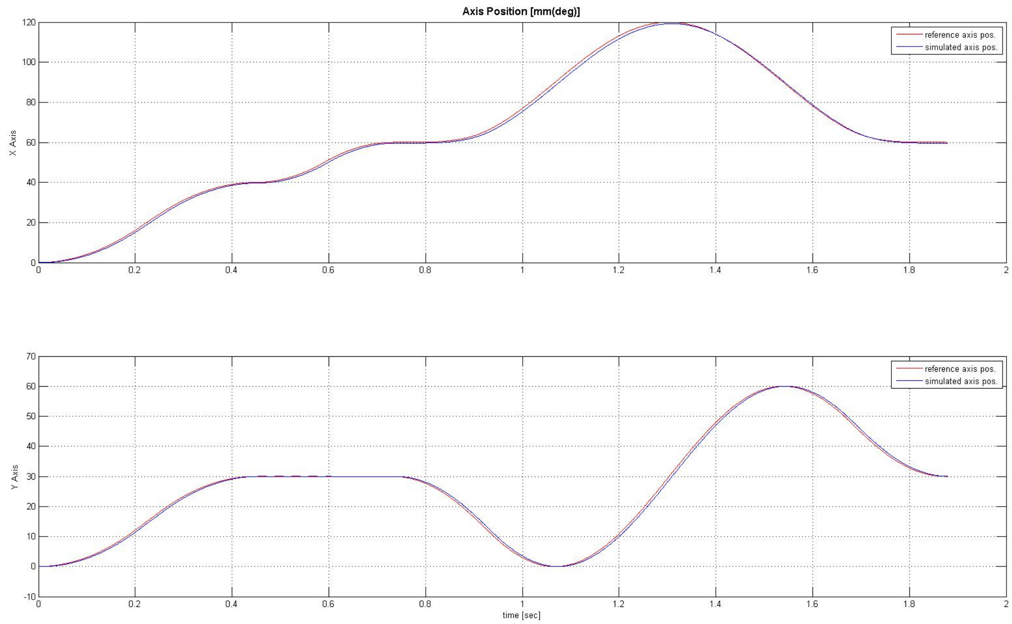

Figure 2: X and Y Axis Position Graphs for PPC

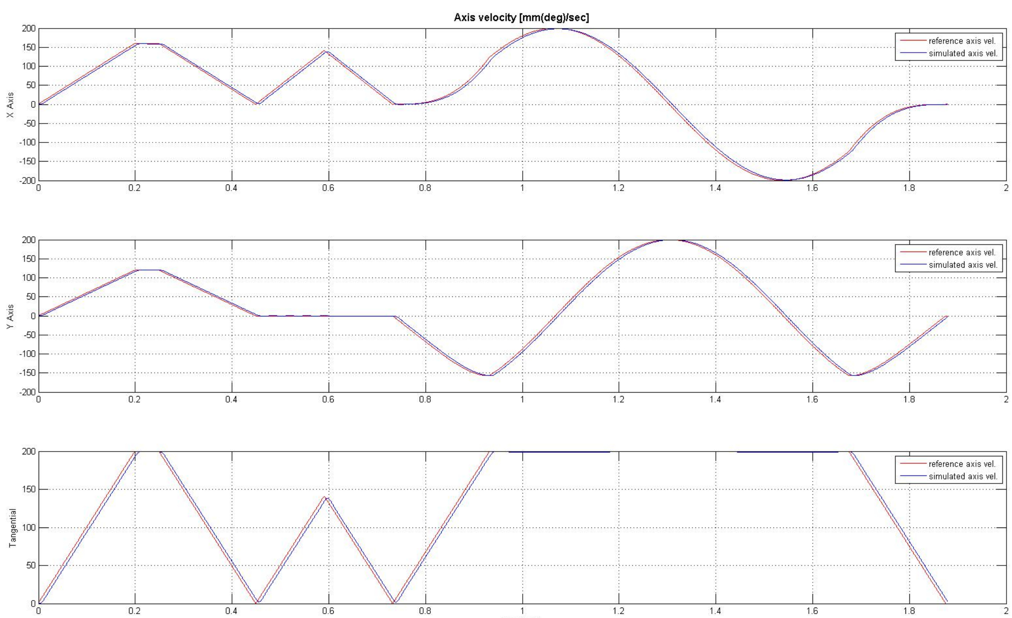

Figure 3: X and Y Axis Velocity Graphs for PPC

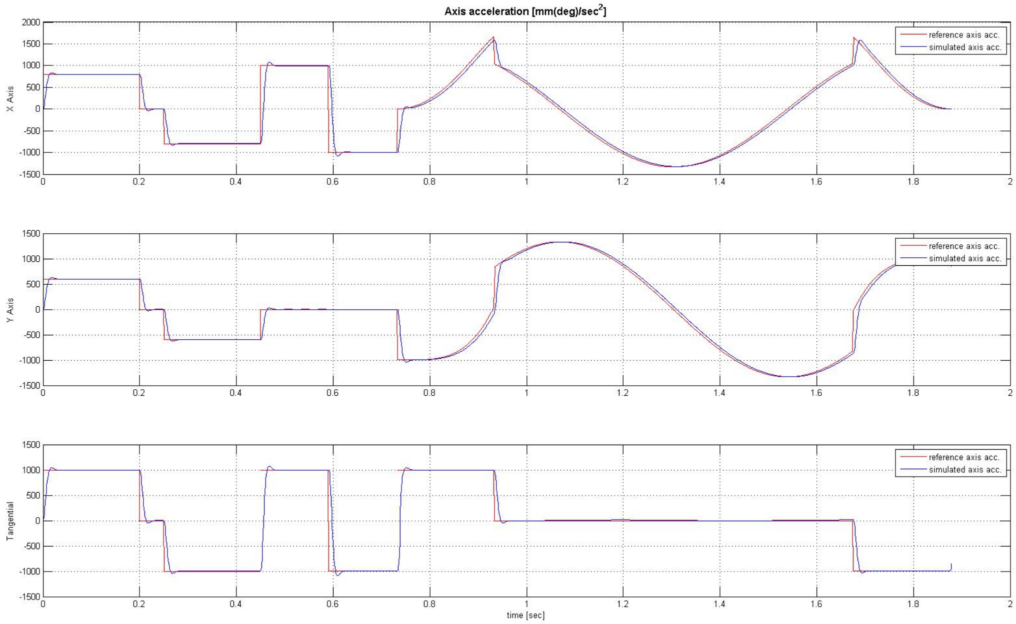

Figure 4: X, Y, and Tangential Acceleration for PPC

The position graphs have very similar behaviour between the X and the Y axis – the reference trace leads the simulated trace everywhere. This is true for the velocity curves too – the simulation curve appears shifted to the right of the reference curve by a few milliseconds.

The acceleration traces also demonstrate this delayed behaviour. Even more interesting is the overshoot that is visible in this plot but was not in the earlier plots. Especially in the regions where acceleration steps from one value to the next, the simulated trace is seen to rise at a non-instantatnious rate, to overshoot, and to correct to the final value after several parts of a second. This delayed response explains the offset in velocity and position. The overshoot is expected as the damping value used in the design of the poles was $\zeta = 0.7$, so the system was underdamped.

4 Pole Placement Controller Errors

Tracking error is the distance between points along a reference trajectory and the resulting points in the actual, or simulated trajectory. Due to the time delay noted in the position graphs, there is likely to be some notable tracking error associated with the pole placement controller. The tracking error from the simulation is shown in Figure 5. For comparison, the X and Y axis tracking error for the low and high bandwidth controllers from Lab 3 are shown in Figures 6 and 7.

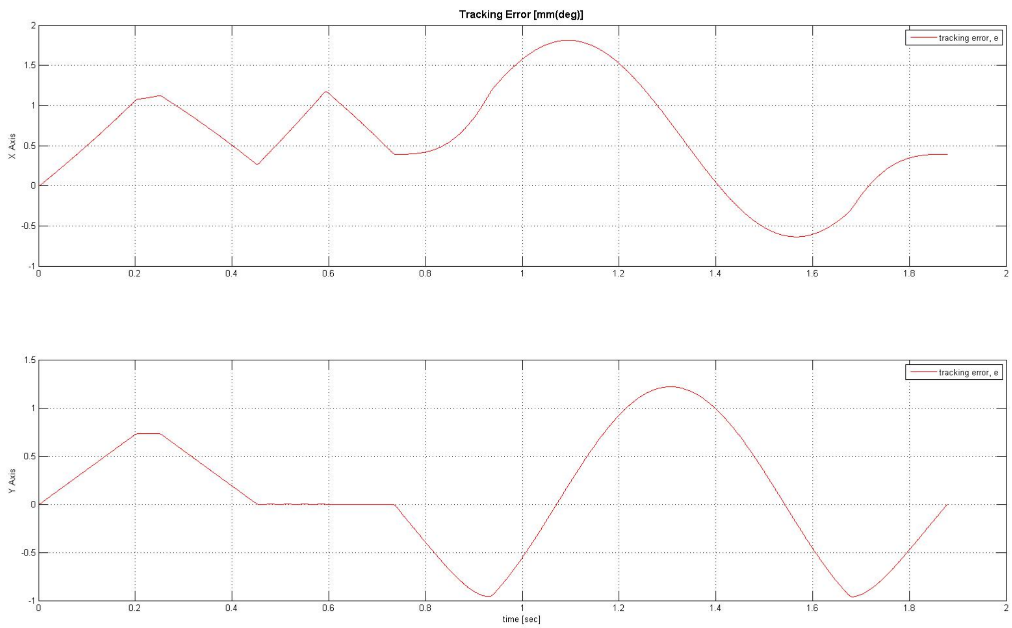

Figure 5: X and Y Axis Tracking Error of PPC

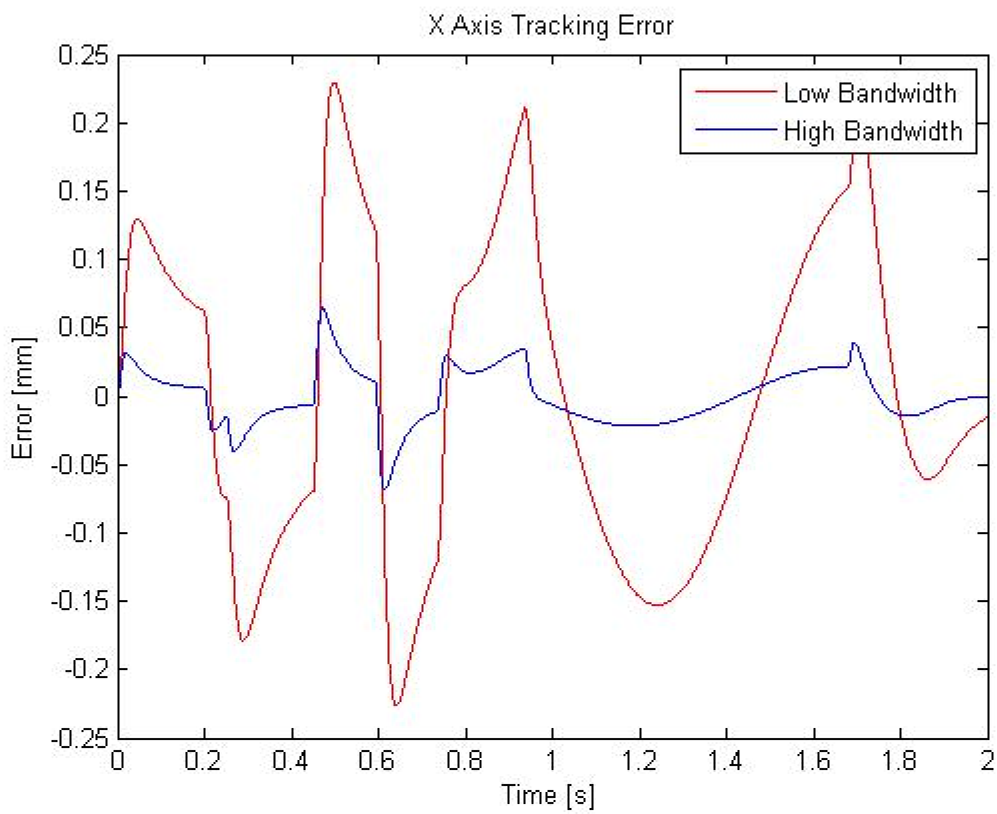

Figure 6: X Axis Tracking Error for LB and HB Controllers

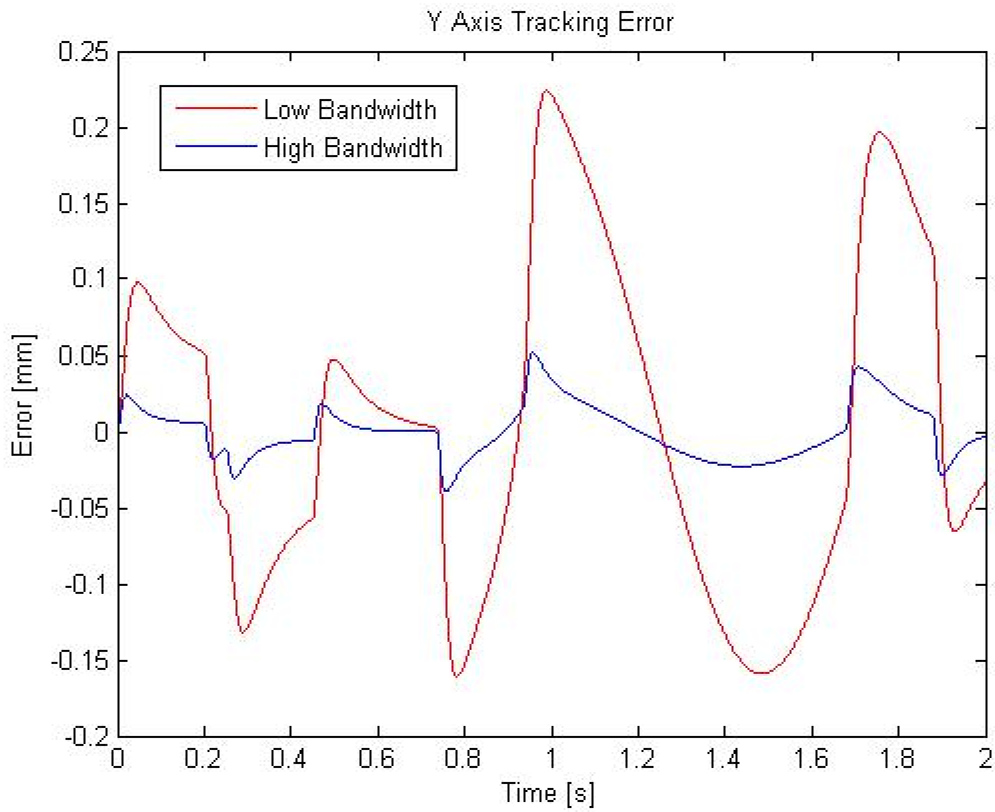

Figure 7: Y Axis Tracking Error for LB and HB Controllers

It is interesting to note how very different these tracking error plots are. Unlike the LB and HB traces, which swap between positive and negative tracking error often throughout the trace, the PPC tracking error plots only change sign when the direction of motion is reversed (during the circle). Additionally, the magnitude of tracking error from the PPC is an order of magnitude larger than that of the LB and HB controllers.

Additionally, the tracking errors follow the general shape of the X and Y velocity traces. Because the position traces follow the same shape but at a delay, the distance between the points increases linearly through changing velocity regions but "catch up" when the velocity is constant.

The X axis PPC Tracking error appears to have some error accumulation, as all the points that were on the Y axis in the velocity plot steadily float upwards as time progresses. This is likely due to a setting neglected in VCNC, and the general shape and magnitude is still acceptable for analysis.

Finally, the contouring error was plotted for the PPC (Figure 8). Select contouring error values from the LB, HB, and PP controllers are shown in Table

1.

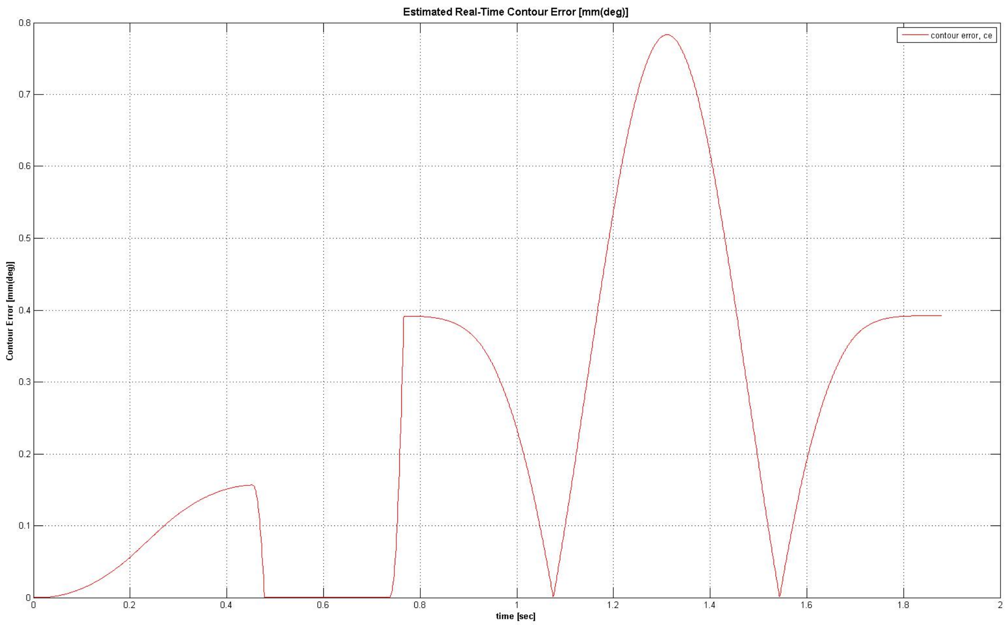

Figure 8: Contouring Error for PPC

Table 1: Contour Errors for LB, HB, and PP Controllers

	LB	HB	PP
R1	0.006 mm;	0.0006 mm;	0.06 mm
R2	0.06 mm	0.005 mm	0.16 mm
R3	0.12 mm	0.013 mm	0.39 mm
R4	0.15 mm	0.016 mm	0.22 mm

Location R1 occurs near 0.2s, R2 occurs near 0.4s, R3 occurs near 0.8s, and R4 occurs near 1.65s.

The contouring errors from the PPC were much larger than those from the LB and HB controllers. The largest PPC contouring error is approximately 0.78 mm, and occurs at the far right hand side of the circle. Depending on the thickness of the instrument tracing this path, this is discrepancy may be visible by eye.

5 CONCLUSION

In conclusion, of the three controllers compared here, the PPC yields the least desirable tracking and contouring errors. Since the angular frequency associated with the circle drawn here is 6.3 rad/s, which is much less than the frequency of 251.3 rad/s that was used when designing this controller, the errors from the PPC must be due primarily to the damping of $\zeta = 0.7$ assigned to the controller. This damping contributes to the slower rise time and overshoot, which is what causes the simulated PPC trace to lag behind the reference trace.